Test cases for 10 case studies:

Case study1:

```
enter classes attended: 10
enter classes conducted: 15
required 75% attendance
your attendance is 66.67
you need to attend 5 upcoming classes to aquire required attendance
enter classes attended: 14
enter classes conducted: 20
required 75% attendance
your attendance is 70.00
you need to attend 4 upcoming classes to aquire required attendance
```

Case study2:

```
Enter number of patients: 2
Enter details for Patient 1
Enter patient name: mustafavali
Enter consultation charges: 2540
Enter diagnostic charges: 3210
Enter treatment charges: 4321
Enter details for Patient 2
Enter patient name: karthik
Enter consultation charges: 4321
Enter diagnostic charges: 2131
Enter treatment charges: 1009
Patient Name: mustafavali
Total Charges: Rs. 10071.0
Concession: Rs. 2014.1999999999998
Final Bill: Rs. 8056.8
Classification: Medium Billing
Patient Name: karthik
Total Charges: Rs. 7461.0
Concession: Rs. 746.1
Final Bill: Rs. 6714.9
Classification: Medium Billing
Total patients processed: 2
```

```
Enter number of patients: 2
Enter details for Patient 1
Enter patient name: mustafavali
Enter consultation charges: 1000
Enter diagnostic charges: 2000
Enter treatment charges: 3000
Enter details for Patient 2
Enter patient name: karthik
Enter consultation charges: 2000
Enter diagnostic charges: 1000
Enter treatment charges: 3500
Patient Name: mustafavali
Total Charges: Rs. 6000.0
Concession: Rs. 600.0
Final Bill: Rs. 5400.0
Classification: Medium Billing
Patient Name: karthik
Total Charges: Rs. 6500.0
Concession: Rs. 650.0
Final Bill: Rs. 5850.0
Classification: Medium Billing
Total patients processed: 2
```

Case study3:

```
available actions: take , return
what is your request: take
what book do you want: book4
return within 5 days!
```

```
available actions: take , return
what is your request: return
what book are you returning: book1
for how many days did you take this book?: 7
you need to pay 150
```

Case study4:

```
Enter your shopping budget: Rs.5000
Enter number of products: 2
Enter product name: shirt
Enter price: ₹620
Enter quantity: 2
Enter product name: pants
Enter price: ₹700
Enter quantity: 3
shirt - Rs. 620.0 x 2 = Rs. 1240.0
pants - Rs. 700.0 x 3 = Rs. 2100.0
Total bill: Rs. 3340.0
Shopping is within your budget.
```

```
Enter your shopping budget: Rs.10000
Enter number of products: 5
Enter product name: shirt
Enter price: ₹500
Enter quantity: 4
Enter product name: pants
Enter price: ₹700
Enter quantity: 3
Enter product name: bags
Enter price: ₹1400
Enter quantity: 2
Enter product name: shoe
Enter price: ₹600
Enter quantity: 1
Enter product name: watch
Enter price: ₹1000
Enter quantity: 1
shirt - Rs. 500.0 x 4 = Rs. 2000.0
pants - Rs. 700.0 x 3 = Rs. 2100.0
bags - Rs. 1400.0 x 2 = Rs. 2800.0
shoe - Rs. 600.0 x 1 = Rs. 600.0
watch - Rs. 1000.0 x 1 = Rs. 1000.0
Total bill: Rs. 8500.0
Shopping is within your budget.
```

Case study5:

```
Enter your name: karthik mohan chandra
Enter your marks percentage: 90
Enter your attendance percentage: 80
Enter your income: 10000
```

```
Enter your name: mustafavali
Enter your marks percentage: 80
Enter your attendance percentage: 70
Enter your income: 70000
the student can apply for scholarship under Need-Based scholarship
```

Case study6:

```
enter password: fghjk
password is short
enter password: cfghjklh
password is short
enter password: ahddbCFGV2332##
your password is strong
```

```
enter password: Asdf2322
password is short
enter password: A@#$adjdns
password doesn't have numbers
enter password: Ajefsj234@!
your password is strong
```

Case study7:

```
Enter number of households: 3
Household 1
Enter household name: mustafa
Enter electricity units consumed: 700
Household 2
Enter household name: karthik mohan chandra
Enter electricity units consumed: 500
Household 3
Enter household name: koushik
Enter electricity units consumed: 300
Household: mustafa
Units consumed: 700.0
Energy charge: Rs. 4900.0
Fixed charge: Rs. 50
Total bill: Rs. 4950.0
Household: karthik mohan chandra
Units consumed: 500.0
Energy charge: Rs. 3500.0
Fixed charge: Rs. 50
Total bill: Rs. 3550.0
Household: koushik
Units consumed: 300.0
Energy charge: Rs. 1500.0
Fixed charge: Rs. 50
Total bill: Rs. 1550.0
Highest bill: mustafa
Amount: Rs. 4950.0
```

```
Enter number of households: 2
Household 1
Enter household name: mustafavali
Enter electricity units consumed: 223
Household 2
Enter household name: srinivas
Enter electricity units consumed: 305
Household: mustafavali
Units consumed: 223.0
Energy charge: Rs. 1115.0
Fixed charge: Rs. 50
Total bill: Rs. 1165.0
Household: srinivas
Units consumed: 305.0
Energy charge: Rs. 2135.0
Fixed charge: Rs. 50
Total bill: Rs. 2185.0
Highest bill: srinivas
Amount: Rs. 2185.0
```

Case study8:

```
available items are:
fried rice
noodles
chicken rice
additional:
eggs
chicken
what you want to order: fried rice
specialization?(0 if no need): chicken
do you want next order('no' if dont need): yes
what you want to order: chicken rice
specialization?(0 if no need): egg
do you want next order('no' if dont need): yes
what you want to order: noodles
specialization?(0 if no need): no
do you want next order('no' if dont need): no
available items are:
fried rice
noodles
chicken rice
additional:
eggs
chicken
what you want to order: fried rice
specialization?(0 if no need): chicken
do you want next order('no' if dont need): yes
what you want to order: noodles
specialization?(0 if no need): egg
do you want next order('no' if dont need): no
```

Case study9:

```
available slots for vehicles ['car', 'bike', 'truck', 'bicycle']
enter your parking vehicle: truck
enter how many hours(0 if less than 1 hour): 8
your total fee is 859
available slots for vehicles ['car', 'bike', 'truck', 'bicycle']
enter your parking vehicle: car
enter how many hours(0 if less than 1 hour): 5
your total fee is 248
```

Case study 10:

```
Enter how many students: 2
Enter your name: mustafa
Enter your marks percentage: 85
Enter your attendance percentage: 70
Enter your income: 70000
Enter your name: koushik
Enter your marks percentage: 80
Enter your attendance percentage: 80
Enter your income: 10000
The eligible students are:
mustafa with scholarship of 70%
koushik with scholarship of 70%
```

```
Enter how many students: 2
Enter your name: mustafavali
Enter your marks percentage: 70
Enter your attendance percentage: 75
Enter your income: 70000
Enter your name: karthik
Enter your marks percentage: 80
Enter your attendance percentage: 60
Enter your income: 80000
The eligible students are:
mustafavali with scholarship of 70%
karthik with scholarship of 30%
```